SUBHRAJIT MISHRA

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EDUCATION

University of California San Diego | *M.S. in Computer Science and Engineering* | San Diego, CA *Sep 2026 – Expected 03/28*

Relevant Courses (Ongoing): Probabilistic Reasoning in AI, Parallel Computation, Introduction to Robotics, Unsupervised Learning (audit)

Research Interests: Efficient Deep Learning, Computer Vision, Multimodal Representation Learning, Neural Compression, ML Systems, Robot Vision

Indian Institute of Technology Kanpur | *B.Tech. in Electrical Engineering* | Kanpur, India *Jul 2020 – May 2024*

CPI: 9.2/10 | Minor: Computer Science: Machine Learning | Academic Excellence Award (2021, 2023, 2024)

Relevant Courses: Deep Learning, RL, Linear Algebra, Stochastic processes, Data Structures and Algorithms, Signal Processing, Computer Organization

RESEARCH AND PROFESSIONAL EXPERIENCE

Samsung Research | *Senior Engineer (Research), Next-Generation Video Compression* | Bengaluru, India *Jul 2024 – Aug 2026*

- **AI in-loop filter for H.267/post-VVC:** Designed a hardware-integrable neural network (<500 FLOPs/pixel) combining artifact-aware handcrafted filters with custom attention blocks to suppress compression artifacts, delivering >3% BD-BR loss over existing standard
- **NN-based intra prediction:** Led the design of a unified, configurable CNN that predicts coding units of varying sizes via custom-built preprocessing & postprocessing blocks at ∼1 KFLOPs/pixel with state-of-the-art BD-BR gains, replacing seven separate model variants
- **Patents:** Inventor on **8 filed patents** in neural video coding (intra prediction, in-loop filtering), including two as first inventor
- **Advanced Professional Video (APV):** Contributed core algorithms to APV, an open-source, perceptually lossless professional video codec; integrated it on Samsung flagship devices and Adobe Premiere Pro for the NAB demo:
- **Performance:** Accelerated decoder speeds of APV and MPEG5-EVC with SIMD intrinsics (ARM NEON, SSE, AVX), by ∼60–70%
- **Achievements:** Received **Star IP Award** for patent contributions, best **Research to Development** Award Q3, multiple **spot awards**

EVERSANA | *Intern, Digital and Data Solutions* | Mumbai, India *May 2023 – Jul 2023*

- Developed a stakeholder "typing tool" that classifies physicians and hospitals into behavioral cohorts and surfaces persona-based insights to raise **Willingness-to-Recommend** (WTR) and **Intent-to-Pay** (ITP) for a client's medical consumables
- Identified WTR/ITP **drivers** and cohort-specific **interventions**; delivered a **client-ready proposal** and **dashboard demo**

RELEVANT PATENTS (FILED)

- **AI-based pixel-block prediction for video codecs** – unified, hardware-efficient intra-prediction network for variable-size coding units
- **Adaptive visual artifact removal using AI in-loop filters** – content-based partitioning of a Mixture of Experts for NN loop filtering
- **Dynamic AI-based multiplexing of NN and classical loop filters** – scalar & patch-level multiplexing to remove compression artifacts
- **On-the-fly kernel adaptation using affine transform in AI loop filtering** – QP-conditioned routing and dynamic weight updates
- **Dynamic local patch modulation in AI in-loop filtering** – spatial-attention-inspired weighting of a patch's local neighborhood

PROJECTS

Multimodal Contrastive Learning for Medical Diagnosis | *B.Tech. Project, Dr. Tushar Sandhan* | *Dec 2023 – May 2024*

- Created a generalizable SimCLR-based **multimodal contrastive learning** framework (InfoNCE loss) fusing imaging (X-ray, MRI, CT) and tabular patient data to diagnose myocardial infarction and coronary artery disease
- Extended it to **medical-equipment classification** (FDA/CDSCO classes) from images and text descriptions, benchmarking against a unimodal BERT text baseline

Deep Reinforcement Learning for Atari | *Advisor: Dr. Subrahmanya Swamy Peruru* | *Mar 2024 – Apr 2024*

- Implemented **DQN** and **Double DQN** agents for CartPole and Atari Breakout/Pong, beating the baseline agent and reaching **maximum-reward convergence**; extended to an LSTM-based **DRQN** matching published results under **partial observability**

UAV-Guided UGV Navigation | *Inter IIT Tech Meet 10.0, DRDO Challenge – Silver Medal* | *Feb 2022 – Mar 2022*

- **Segmented drone camera frames** to locate the road and vehicle orientation; combined **PID** velocity and **pure pursuit** controllers for throttle and steering; enabled **autonomous takeoff** and mean-path extraction in drone for **<5m** deviation of UGV on road

Introduction to Machine Learning | *Advisor: Dr. Purushottam Kar* | *Jan 2023 – Apr 2023*

- Broke simple and advanced **XORRO PUFs** with >99% accuracy using **linear models** (Linear SVC, Logistic Regression)
- Built a **decision-tree** agent for a Hangman–Wordle hybrid with 100% accuracy at an average of **3.99** queries (**max-info-gain** splits)

TECHNICAL SKILLS

- **Programming:** C++, Python, C, MATLAB • **Tools:** Linux, Git, Bash, Visual Studio, ROS, Gazebo, LaTeX
- **Machine Learning / Deep Learning / Computer Vision:** PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas
- **Performance & Video Coding:** SIMD intrinsics (ARM NEON, SSE, AVX2), H.266, MPEG-5 EVC, APV, FFmpeg, neural codecs

LEADERSHIP AND MENTORING

- **Team Head, Aerial Robotics, IITK:** led a 30-member team in (inter)national competitions; recruited and mentored 25 juniors *'21 – '23*
- **Project Mentor, Stamatics Society:** mentored ∼100 undergraduates in image processing for computer vision *2023 – 2023*